

Chapter # 4 - E - Safety and Health and Safety

Specimen 2017

- 8 The internet is used by many individuals in the world today.
- (a) There are a number of malware issues associated with using the internet. Other than viruses, describe **three** of these issues.

Answer:

8(a)	Award a maximum of 2 marks per answer, up to a maximum of three types of malware. A Trojan Horse is generally a non-self-replicating program containing malicious code that, when executed, carries out actions, typically causing loss of data Malicious programs are classified as Trojans if they do not attempt to inject themselves into other files... ...or otherwise propagate themselves A worm is a computer program that replicates itself in order to spread to other computers It uses a computer network to spread itself Unlike a computer virus, it does not need to attach itself to an existing program Spyware is software that aids in gathering information about a person or organisation without their knowledge It may send such information to another entity without the consumer's consent, asserts control over a computer without the consumer's knowledge It detects key presses of the computer user	6
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Question 8, Part (a)**Answer (Continued):**

Adware is a form of spyware that collects information about the user in order to display advertisements in the Web browser
Some applications that contain adware track the user's internet surfing habits in order to serve ads related to them

A rootkit is a type of software designed to hide the fact that an operating system has been compromised...
...sometimes by replacing vital executables
Rootkits allow viruses and malware to "hide in plain sight" by disguising as necessary files that antivirus software will overlook
Rootkits themselves are not harmful – they are simply used to hide malware, bots and worms

Malicious bots are self-propagating malware designed to infect a host and connect back to a central server or servers
The servers act as a command and control (C&C) centre for an entire network of compromised devices
Hackers can launch flood-type attacks against their target(s)
In addition to the worm-like ability to self-propagate, bots can act as spyware

Ransomware is a type of malware which restricts access to the computer system that it infects
It demands a ransom paid to the creator(s) of the malware in order for the restriction to be removed
Some forms of ransomware encrypt files on the system's hard drive

June 2017 – P11 & P12

- 8 Li is very concerned that one day he could be a victim of a scam to take money from his bank account. He is particularly worried about smishing and pharming.

Describe what is meant by

- (a) smishing.

Answer:

8(a)	Three from: Involves using cell phone text/SMS messages to persuade people to divulge their personal information The text message may include a website URL, inviting the receiver to go to the site... ...looks just like the actual bank's website but is a fake website The text message may include a telephone number that connects to an automated voice response system/person asking for bank details The smishing message usually contains something that demands the target's immediate attention The website then asks them to enter their personal/financial information.	3
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- (b) pharming.

Answer:

8(b)	Three from: Installs a piece of malicious software/code on customer's computer Creates a fake website which looks like the actual bank's website Fraudster redirects genuine website's traffic to own website... ...customer is now sending personal details to fraudster's website unknowingly.	3
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June 2017 – P13

6 In a recent BBC news bulletin it was reported:

The BBC has obtained exclusive figures from the financial ombudsman that show there have been nearly 100 complaints about “vishing” in the past three months.

Describe what is meant by:

(i) phishing.

Answer:

6(i)	Phishing Three from: Involves using email to persuade people to divulge their personal information The email may include a website URL, inviting the receiver to go to the site... ...looks just like the actual bank's website but is a fake website The email may ask customer to reply with their bank details The email usually contains something that demands the target's immediate attention Examples include "We confirm that you have signed up to our service. You will be charged \$2 a day unless you cancel your order on this URL: ..." The website then asks them to enter their personal/financial information.	3
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(ii) vishing.

Answer:

6(ii)	Vishing Three from: Victim receives a phone call to persuade him to divulge his personal information Claims to be a representative of the bank and says... ...that there has been some fraudulent activity on the customer's account Often involves having the customer phone the bank Fraudster's computer redirects phone call to own phone Gives customer their bank details to get their confidence Gets customer to log on to bank's site and enter their details to transfer money to their new account which has been set up This account is actually the fraudster's own bank account.	3
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November 2017 – P11 & P12

- 2 Many schools want IT departments to introduce lessons emphasising aspects of e-safety.

Tick **four** of the following statements which offer good advice.

	✓
It is alright to send an email to a person you do not know as it does not matter if they find out your email address.	
You should always attach a photograph of yourself to an email so they know who you are.	
You should only use websites recommended by teachers.	
It is acceptable to post a photograph of your school on a social media site as so many students go there.	
You should always think twice before opening an email from an unknown person as it might be spam.	
It is okay, for the first time, to meet a person you only know from the internet on your own.	
You should know how to block and report unwanted users from social media sites.	
It is acceptable, if you trust them, to give someone you have only met on the internet your name and address.	
When using instant messaging it is okay to use bad language if you are among friends.	
When playing games on the internet you should never use your real name.	

[4]

Question 2**Answer:**

2	It is okay to send an email to a person you do not know as it does not matter if they find out your email address. You should always attach a photograph of yourself to an email so they know who you are. You should only use websites recommended by teachers. It is acceptable to post a photograph of your school on a social media site as so many students go there. You should always think twice before opening an email from an unknown person as it might be spam. It is okay, for the first time, to meet a person you only know from the internet on your own. You should know how to block and report unwanted users from social media sites. It is acceptable, if you trust them, to give someone you have only met on the internet your name and address. When using instant messaging it is okay to use bad language if you are among friends. When playing games on the internet you should never use your real name.	✓ ✓ ✓ ✓	4
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June 2018 – P13

- 2 Tick the **four** most accurate statements referring to protecting the confidentiality of personal data.

	✓
You should use different passwords and PINs for different accounts.	
Storing personal data on removable media only and locking them away when not in use, is a good form of protection.	
Encryption prevents data from being accessed.	
A firewall always prevents unauthorised users from accessing confidential data.	
Installing spyware helps protect confidential data from unauthorised access.	
Banks are allowed to share personal data of customers with other customers.	
Confidential personal data is usually stored on DVD ROMs as this prevents it from being accessed.	
TLS is a cryptographic protocol used to protect personal data transmitted across a network.	
Users should always log off immediately after using a site where personal data has been typed in.	
Using public computers to access personal information is good practice as they usually have plenty of security.	

[4]

Question 2**Answer:**

2	You should use different passwords and PINs for different accounts.	✓	4
	Storing personal data on removable media only and locking them away when not in use, is a good form of protection.	✓	
	Encryption prevents data from being accessed.		
	A firewall always prevents unauthorised users from accessing confidential data.		
	Installing spyware helps protect confidential data from unauthorised access.		
	Banks are allowed to share personal data of customers with other customers.		
	Confidential personal data is usually stored on DVD ROMs as this prevents it from being accessed.		
	TLS is a cryptographic protocol used to protect personal data transmitted across a network.	✓	
	Users should always log off immediately after using a site where personal data has been typed in.	✓	
	Using public computers to access personal information is good practice as they usually have plenty of security.		

November 2018 – P11 & P13

- 7 Describe, in detail, **three** different safety (apart from e-safety) issues that can arise from the use of IT. For each issue give **one** method of helping to prevent it. Each method must be different.

Answer:

7	Three matched pairs <i>from</i>: Overloading sockets causing overheating and therefore fire CO₂ fire extinguisher in the room/separate sockets for each plug Water spilt on to live wires/handling bare wires can cause electrocution Do not allow food and drink into the computer room/keep computers away from water supply/ensure regular inspection by electrician/ensure all wires are properly insulated Trailing cables can cause users to trip up and injure themselves Ensure proper trunking is in place/keep cables under carpets/use Wi-Fi Heavy objects can fall off tables and cause injury Ensure sturdy desks or tables are used / heavy objects are placed in the centre of tables. 1 mark for description of issue and 1 for prevention.	6
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November 2018 – P12

3 One of the dangers of accessing the internet is the increased risk of malware attacks.

Describe the distinctive features of each of the following types of malware.

Answer:

3	Trojan Horse: Two from: A malicious computer program which is used to hack into a computer by misleading users of its true intent/disguises itself as authentic software (unlike computer viruses and worms) Trojans generally do not attempt to infect other files They do not attempt to self-replicate themselves Worm: Two from: A standalone malicious computer program that replicates itself in a computer system Many copies are then sent to other computers Worms almost always cause some harm/slowing down to the network even if only by consuming bandwidth Spyware: Two from: Software designed to collect information about a computer user's activities Data such as passwords is passed to a remote server/attacker/hacker (without the user's knowledge) A keylogger is a type of spyware (which collects a record of the user's key strokes) Adware: Two from: Automatically generates advertisements in order to gain revenue for its author Advertisements may appear in the user interface/screen shown to the user by the software Examines which internet sites visited and presents advertising according to the types of goods/services featured on these/usually the type of goods/adverts the user is interested in.	8
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March 2018 – P12

- 5 Describe, in detail, **four** health issues that can arise from the use of IT. For each issue give **one** method of preventing it. Each prevention must be different.

Answer:

5	Four matched pairs from: RSI in the fingers/wrists/carpal tunnel syndrome from continual typing/clicking of mouse Take a break every hour/take regular breaks/use wrist rests/use ergonomic keyboard Cubital Tunnel syndrome from prolonged use of mobile phones keeping elbow bent Use of headphones and microphone/use of Bluetooth headset/earpiece Headaches/eyesight problems from eyestrain caused by staring at the monitor for long periods Take regular breaks / focus on distant objects from time to time/use anti-glare screen Upper back/neck problems by continually staring at the monitor from below/above Position monitor at eye height and appropriate distance Lower back pain caused by sitting in same position for long periods Use ergonomic furniture/ sit up straight/use a back support/foot rests.	8
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June 2019 – P11

- 4 Most schools have introduced e-safety lessons because teachers are becoming concerned about the way their students use the internet when not in school. One of their concerns is that students may be revealing personal information about themselves.

Describe the problems associated with this and other problems that arise from students using the internet.

Answer:

4	Six from: - Unauthorised access to personal information - The risk of being subject to grooming by those with whom they make contact on the internet - The sharing / distribution of personal images without an individual's consent or knowledge - Inappropriate communication / contact with others, including strangers - Cyber-bullying by fellow students <i>Other problems</i> - Access to unsuitable video / internet games - An inability to evaluate the quality, accuracy and relevance of information on the internet - Plagiarism and copyright infringement - Illegal downloading of music or video files... ...giving rise to the potential for excessive use which may affect the social and emotional development and learning of the student - Access to illegal, harmful or inappropriate images or other content Must have at least two of each to gain full marks	6
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8 There are a number of ways that fraudsters can obtain money from customers' bank accounts.

(a) Describe how the fraudster would use vishing to obtain a customer's personal data.

Answer:

8(a)	Three from: Fraudster phones the victim pretending it is a call from the bank Fraudster claims that there has been some fraudulent activity on the customer's account/customer has made certain purchases – could they confirm? Often involves getting the customer to phone the bank Fraudster's computer redirects phone call to own phone Gives customer their bank details to get their confidence Gets customer to log on to bank's site and enter their details to transfer money to their new account which has been set up This account is actually the fraudster's own bank account	3
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(b) Describe how the fraudster would use smishing to obtain a customer's personal data.

Answer:

8(b)	Three from: Fraudster sends a cell phone text message to persuade customer to divulge their personal Information... ...inviting the receiver to go to the fraudster's website They include a telephone number in the message that connects to an automated voice response system They include in the message something that demands the target's immediate attention such as... ... "We confirm that you have signed up to our service. You will be charged \$2 a day unless you cancel your order on this URL" The user then goes on to a seemingly legitimate website that asks them to "confirm/enter your personal financial information"	3
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June 2019 – P12

2 Tick the **four** statements referring to malware which are true.

	✓
A virus corrupts files but does not delete them	
Ransomware continually deletes files until a ransom is paid	
One purpose of releasing a worm is to occupy as much bandwidth in a network as possible	
Ransomware is often initiated by means of a trojan horse	
A trojan horse is a type of malware within an email attachment that is downloaded without the user understanding the consequences	
Spyware is a type of software that removes key loggers from a hard disk	
Adware is additional software attached to a virus when it is downloaded	
Malicious bots are never used to instigate a SPAM attack	
A rootkit enables a hacker to gain administrator rights in a victim's computer	
A trojan horse just enables a virus to be transmitted. It never deletes data itself	

[4]

Question 2**Answer:**

2	A virus corrupts files but does not delete them		4
	Ransomware continually deletes files until a ransom is paid		
	One purpose of releasing a worm is to occupy as much bandwidth in a network as possible	✓	
	Ransomware is often initiated by means of a trojan horse	✓	
	A trojan horse is a type of malware within an e-mail attachment that is downloaded without the user understanding the consequences	✓	
	Spyware is a type of software that removes key loggers from a hard disk		
	Adware is additional software attached to a virus when it is downloaded		
	Malicious bots are never used to instigate a SPAM attack		
	A rootkit enables a hacker to gain administrator rights in a victim's computer	✓	
	A trojan horse just enables a virus to be transmitted as it never deletes data itself		

November 2019 – P11 & P13

- 2 Tick the **four** most accurate statements regarding the gathering of personal data by unauthorised persons.

	✓
Pharming involves sending a website URL, inviting the receiver to go to the website	
Smishing involves using text messages to persuade people to give out personal information	
Phishing involves installing malicious software on the victim's computer	
Pharming involves creating a fake website that looks like an actual bank's website	
Phishing involves the sending of fake emails in order to advertise the bank	
Vishing involves the fraudster's computer redirecting a customer's phone call to his own phone	
Banks can ask you from time to time to send them your full password to check you are authorised to use that account	
Vishing involves receiving a phone call from a fraudster	
Vishing never involves getting the customer to phone the bank	
Smishing is the same as vishing but never involves the use of a phone	

[4]

Question 2**Answer:**

2	Pharming involves sending a website URL, inviting the receiver to go to the site.		4
	Smishing involves using text messages to persuade people to give out personal information.	✓	
	Phishing involves installing malicious software on the victim's computer.		
	Pharming involves creating a fake website that looks like an actual bank's website.	✓	
	Phishing involves the sending of fake emails in order to advertise the bank.		
	Vishing involves the fraudster's computer redirecting a customer's phone call to his own phone.	✓	
	Banks can ask you from time to time to send them your full password to check you are authorised to use that account.		
	Vishing involves receiving a phone call from a fraudster.	✓	
	Vishing never involves getting the customer to phone the bank.		
	Smishing is the same as vishing but never involves the use of a phone.		

March 2019 – P12

- 9 Describe the differences between health issues and safety (apart from e-safety) issues experienced by employees in an IT-based work environment.

Answer:

9	Six from: Health issues arise from <u>long term use</u> of computers Safety issues can result in a <u>sudden</u> accident/can result in serious injury Health issues tend to be RSI/carpal tunnel syndrome, lower back pain, eye strain, headaches, upper back pain, neck pain, shoulder pain (must have at least two) <i>Health issues can be caused by:</i> Max two from: Typing at a keyboard <u>continuously</u> Gripping a mouse and <u>repetitive</u> clicking Sitting in the <u>same position/with wrong posture all day</u> Staring at a computer screen <u>continuously/all day</u> <u>Glare</u> from screen Safety issues tend to be fire hazards, electrocution, personal injury (must have at least two) <i>Safety issues can be caused by:</i> Max two from: Overloading sockets <u>causing overheating</u> Overheating of computers Water spilt on to live wires Handling bare wires, Trailing cables can cause users to trip up Heavy objects falling off tables.	6
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